Functions and Polynomials

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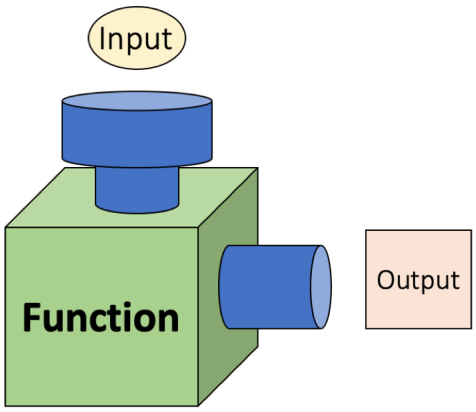
In these notes, we will discuss the fundamental principles of polynomials and functions. We will examine the properties of polynomial equations, root-finding strategies, and analyze the behavior of functions at a tactile and intuitive level.

Functions

Function

A function is a relationship between two sets where one element of the first set corresponds with another element of a second set. Mathematically, a function $f : X \to Y$ maps an input x from the set X to one element $y = f(x)$ from the set Y.

A good analogy is a machine, which turns the input of raw materials and produces an output of a finished product.



Consider the identity function $f(x) = x$ for $f : \mathbb{R} \to \mathbb{R}$ ($\mathbb{R}$ is the set of real numbers). We realize that $f(1) = 1$, $f(2) = 2$, and this function simply returns the exact input. We also note that if $x_1 = x_2$, then $f(x_1) = f(x_2)$ as $f(x_1) = x_1 = x_2 = f(x_2)$. It actually turns out that all functions satisfy this property since one input $x_1 = x_2$ cannot map to two different outputs, so $f(x_1) \neq f(x_2)$ is impossible. In order to rapidly determine whether a curve in the xy plane is a function, we perform the vertical line. If there is a vertical line that intersects the curve at more than one point, the curve is *not* a function.

Each function $f : X \to Y$ has a domain and range. The domain X is the set of all possible input values and the range Y is the set of all possible output values. Typically, the domain is shown on the x-axis and the range is illustrated on the y-axis. For instance, we can represent a function g the set of ordered pairs $\{(0, 0), (1, 2), (2, 4), (3, 6)\}$, where each ordered pair (x, y) represents an x and y-coordinate. The domain is the set of all possible inputs x, which is $\{0, 1, 2, 3\}$. The range is the set of all possible outputs y, which turns out to be $\{0, 2, 4, 6\}$.

Functions are, of course, conceptually applicable outside the mathematical domain. An Instagram filter $f : X \to Y$ represents a mapping of a regular photograph x to a filtered photo y, or $f(x) = y$, as depicted below. In computer science, this is called an image-to-image translation task.

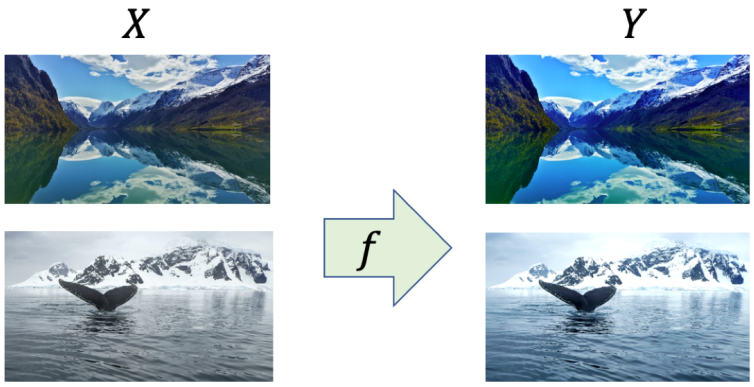


Figure 1: Here, the filter f is Instagram Clarendon.

Example 1

Determine whether each of the following are functions.

1. $y = x^2$
2. $x^2 + y^2 = 1$
3. $y^4 = x + 1$

Solution: In the first equation, we note that for any value of x, there is only one value of y that corresponds with it, which is x^2. Thus, (1) $y = x^2$ is a function. In the second equation, we can rewrite the equation as $y^2 = 1 - x^2$, or $y = \pm\sqrt{1 - x^2}$. Since one value of x, say 0, can correspond with multiple values of y—in this case, ± 1—then (2) is not a function. Finally, we note that $(-y)^4 = y^4 = x + 1$. Thus, multiple values of y can correspond to the same value of x, meaning that (3) is not a function. Thus, only $\boxed{(1)}$ is a function.

Example 2

A function f has domain [0,2] and range [0,1]. (The notation [a,b] denotes $\{x : a \leq x \leq b\}$.) What are the domain and range, respectively, of the function g defined by $g(x) = 1 - f(x+1)$? (AMC12A 2008)

Solution: We first consider the intermediate function $f(x+1)$, which has a different domain as $f(x)$ since the input must be between 0 and 2. Thus, $0 \leq x + 1 \leq 2$, meaning that $-1 \leq x \leq 1$. Furthermore, $g(x) = 1 - f(x+1)$ has the same domain as $f(x+1)$ since the x is entered through the function f so $\boxed{\text{the domain of } g(x) \text{ is } [-1, 1]}$.

The range is slightly more interesting in that $f(x+1)$ and $f(x)$ actually have the same range since both functions are receiving an identical set of inputs (the domain for $f(x+1)$ is slanted 1 unit from $[0, 2]$ to $[-1, 1]$ to maintain the same inputs). Thus, the $0 \leq f(x+1) \leq 1$. We note that since $g(x) = 1 - f(x+1)$, g is largest when $f(x+1)$ is small and smallest when $f(x+1)$ is a large number. Thus, $g(x)$ ranges from $\min(g) = 1 - \max(f(x+1)) = 1 - 2 = -1$ to $\max(g) = 1 - \min(f(x+1)) = 1 - 0 = 1$. This suggests that $\boxed{\text{the range of } g \text{ is } [-1, 1]}$.

Polynomials

Polynomial

A polynomial function is a function f of the form $f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x^1 + a_0$, where $a_0, a_1, ..., a_n$ are constants. We say that this polynomial has degree n, where degree refers to the highest of the degrees of all the non-zero terms. Using summation notation, we can also write f as $\sum_{i=0}^{n} a_i x^i$.

Examples of polynomials include $f(x) = x^2 + 2x - 3$ and $g(x) = x^3 + 3x^2 + 3x + 1$. We can add and subtract polynomials by grouping together like terms.

$$f(x) + g(x) = x^2 + 2x - 3 + x^3 + 3x^2 + 3x + 1 = x^3 + 4x^2 + 5x - 2.$$
$$f(x) - g(x) = x^2 + 2x - 3 - (x^3 + 3x^2 + 3x + 1) = -x^3 - 2x^2 - x - 4.$$

We also multiply polynomials by repeatedly applying the distributive property, or using an extended version of FOIL expansion. For instance, we can multiply $p(x) = x^2 + x - 2$ and $q(x) = x^2 + 2x + 1$ as shown below. Note that each term in p is multiplied once with *every* term in q.

$$\begin{aligned} p(x) * q(x) &= (x^2 + x - 2) * (x^2 + 2x + 1) \\ &= (x^2 * x^2 + x^2 * 2x + x^2 * 1) + (x * x^2 + x * 2x + x * 1) + (-2 * x^2 - 2 * 2x - 2 * 1) \\ &= x^4 + 2x^3 + x^2 + x^3 + 2x^2 + x - 2x^2 - 4x - 2 \\ &= x^4 + 3x^3 + x^2 - 3x - 2. \end{aligned}$$

To divide polynomials, we often use long division, as sketched in the approach below.

$$\begin{array}{r|l} & \quad x^2 \quad 4x \quad 10 \\ \hline x - 2 & x^3 + 2x^2 + 2x + 1 \\ & x^3 - 2x^2 \\ \hline & \quad 4x^2 + 2x \\ & \quad 4x^2 - 8x \\ \hline & \qquad 10x + 1 \\ & \qquad 10x - 20 \\ \hline & \qquad\quad 21 \end{array} \qquad \frac{x^3 + 2x^2 + 2x + 1}{x - 2} = x^2 + 4x + 10 + \frac{21}{x - 2}$$

Figure 2: Long division approach that produced a quotient of $x^2 + 4x + 10$ and remainder of 21.

Composition

We call $+, -, *$, and $\%$ binary operators in the sense that they always involve two numbers: $a + b$, for instance. Another binary operator that is vital to the study of functions is the composition operator used in an expression such as $f \circ g$. This is equivalent to $f(g(x))$. The process for evaluating $f(g(x))$ involves first processing x through g and taking that output and feeding it through f, as seen below. Suppose that $f(x) = 2x + 5$ and $g(x) = x + 1$. Then, $f(g(x)) = f(x + 1) = 2(x + 1) + 5 = 2x + 2 + 5 = 2x + 7$.

Using a similar approach, we can find that $g \circ f = g(f(x)) = g(2x + 5) = (2x + 5) + 1 = 2x + 6$. Note that $f \circ g$ and $g \circ f$ are not necessarily equal.

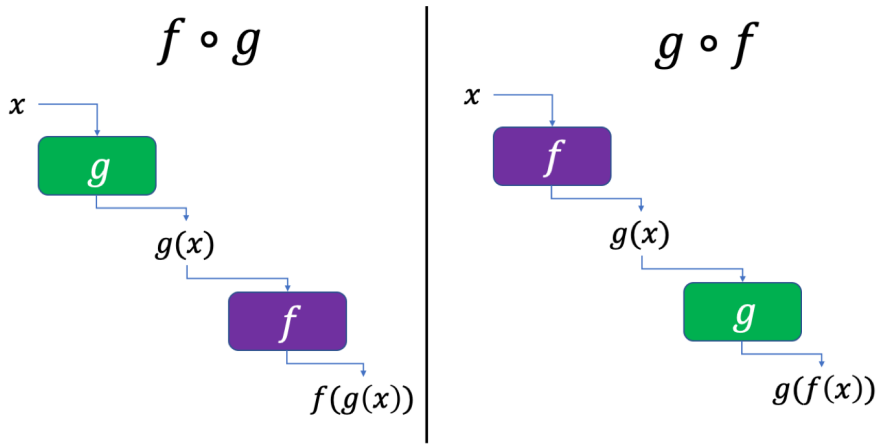


Figure 3: Diagram representation of $f \circ g = f(g(x))$ and $g \circ f = g(f(x))$.

Example 3

Suppose $P(x)$ is a quadratic polynomial with integer coefficients satisfying the identity

$$P(P(x)) - P(x)^2 = x^2 + x + 2016$$

for all real x. What is $P(1)$? (CMIMC 2017)

Since P is a quadratic, it is a degree 2 polynomial and therefore has the form $P(x) = ax^2 + bx + c$ for integer coefficients $a \neq 0$, b, and c. Thus, substituting $P(x)$ into the equation in the problem and expanding produces

$$\begin{aligned} P(P(x)) - P(x)^2 &= P(ax^2 + bx + c) - (ax^2 + bx + c)^2 \\ &= a(ax^2 + bx + c)^2 + b(ax^2 + bx + c) + c - (ax^2 + bx + c)^2 \\ &= (a-1)(ax^2 + bx + c)^2 + b(ax^2 + bx + c) + c. \end{aligned}$$

We know that this expression is equal to $x^2 + x + 2016$, which is a degree 2 polynomial. However, if $a \neq 1$, then $P(P(x)) - P(x)^2$ will have a degree higher than 2 due to the term $(a-1)(ax^2 + bx + c)^2$, which has a degree of 4 unless it is canceled out. Thus, $a = 1$, which enforces

$$\begin{aligned} P(P(x)) - P(x)^2 &= (a-1)(ax^2 + bx + c)^2 + b(ax^2 + bx + c) + c \\ &= b(x^2 + bx + c) + c \\ &= bx^2 + b^2x + (bc + c). \end{aligned}$$

Looking at each of the corresponding coefficients with $x^2 + x + 6$, we note that $b = 1$ and $bc + c = 1 * c + c = 2016$, so $c = 1008$. Thus, $P(x) = x^2 + x + 2016$ and $P(1) = 1^2 + 1 + 1008 = \boxed{1010}$.

Inverse Functions

Inverse

Let f and g be two injective functions. If both of the following two statements are true, we argue that f and g are mathematical inverses of one another. We can then write $g(x) = f^{-1}(x)$ and $f(x) = g^{-1}(x)$.

$$(f \circ g)(x) = f(g(x)) = x$$
$$(g \circ f)(x) = g(f(x)) = x$$

Quite literally, g is performing the inverse of $f(x)$ by reversing the function's action. After applying f to translate x to $f(x)$, applying g will retranslate $f(x)$ back into $g(f(x)) = x$. As a note, it is not true that $f^{-1}(x) = \frac{1}{f(x)}$, which is a common error when dealing with inverse functions.

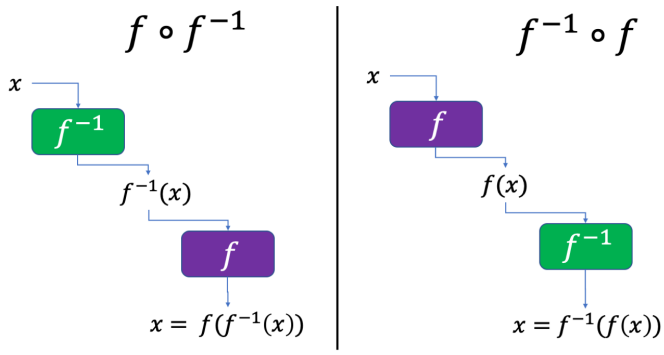


Figure 4: Diagram representation of inverse functions f and $f^{-1}(x)$.

To find the inverse of a particular function procedurally, we first replace $f(x)$ with y. Then, we swap the x and y variables in the equation and solve the equation in terms of y. Finally, we switch out y with $f^{-1}(x)$, which culminates the process.

Example 4

Compute the inverse of the function $f(x) = x + 5$ for $x \in R$.

Solution: We convert the function into $y = x + 5$. Swapping the x and y variables produces $x = y + 5$, which can be rewritten into $y = x - 5$ and $f^{-1}(x) = \boxed{\text{x-5}}$.

An interesting note about inverse functions is that they are reflections of each other across the line $y = x$ when graphed. A few examples for this phenomena are provided below.

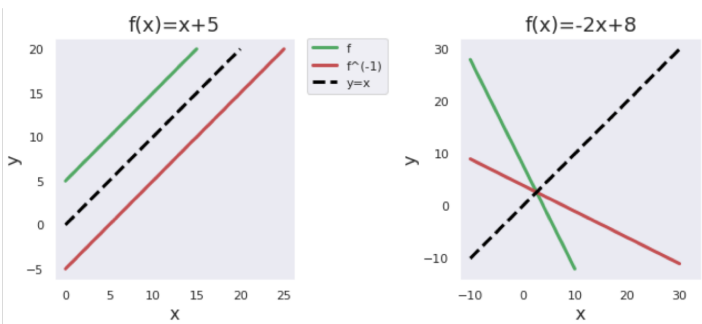


Figure 5: The functions f are in green, the inverses are in red, and $y = x$ is the dashed black line.

We also note that functions had a valid inverse if they pass the horizontal line. Essentially, if each horizontal line passes the graph more than once over the domain, the function is not injective and thus has no valid inverse. For instance, $f(x) = x^2$ violates the horizontal line test since $(-1, f(-1))$ and $(1, f(1))$ occur on the same horizontal line.

The inverse of these functions often include truncated domains where the function passes the horizontal line test. For instance, given a function $f(x) = x^2$, we can snip its domain to the window $[0, \infty]$. Thus, the inverse will be $x = y^2$, which can be formalized into $y = \sqrt{x}$ (as $x \geq 0$) and thus $f(x) = \sqrt{x}$. Similarly, for a more complex quadratic such as $f(x) = x^2 + 2x + 2$, we can rewrite f as $f(x) = (x^2 + 2x + 1) + 1 = (x + 1)^2 + 1$. Thus, to find a valid $f^{-1}(x)$, we truncate the domain of f to $[-1, \infty]$ so that it passes the horizontal line test. Swapping x and y to generate the inverse produces $x = (y + 1)^2 + 1$. Thus, $(y + 1)^2 = x - 1$ and $y + 1 = \sqrt{x - 1}$ (the right hand side must be positive since $x - 1 \geq 0$) and $f^{-1}(x) = y = \sqrt{x - 1} - 1$.

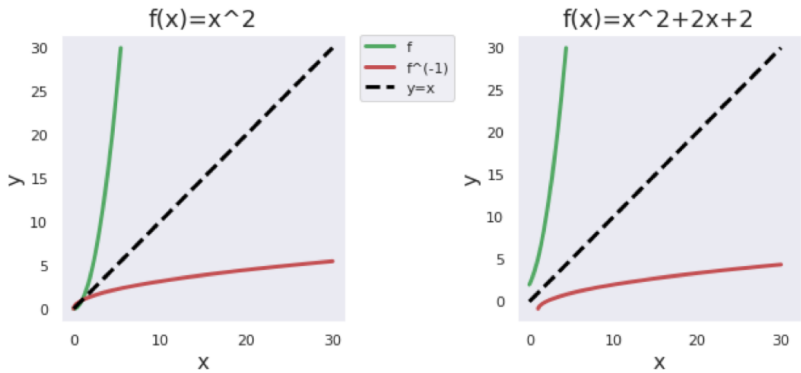


Figure 6: The functions f are in green, the inverses are in red, and $y = x$ is the dashed black line. Note that the domains for each function have been truncated from the typical $[-\infty, \infty]$.

Example 5

If $f(x) = ax + b$ and $f^{-1}(x) = bx + a$ with a and b real, what is the value of $a + b$? (AMC 12B 2004)

Solution: We know that $f(f^{-1}(x)) = x$. Thus, $f(bx + a) = x$, which produces $a(bx + a) + b = x$ and $abx + a^2 + b = x$. By comparing the coefficients, we find that $ab = 1$ and $a^2 + b = 0$. Thus, we have $b = -a^2$ and plugging this into the other equation yields $ab = a(-a^2) = -a^3 = 1$. Since a is real, we have $a = -1$ and consequently $b = -1$. Therefore, $a + b = \boxed{-2}$.

An interesting principle concerning inverses is that when inverting a function, the domain and range actually swap places. The domain of f is the same as the **range** of f^{-1}, and the range of f is identical to the **domain** of f^{-1}. This is because f and f^{-1} are reflections of each other across the graph $y = x$, and such a reflection will swap the x and y variables. Since x-axis is occupied by the domain and the y-axis is occupied by the range, the domain and range swap territories when x and y are switched.

	Domain	Range
$\sin(x)$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	$[-1,1]$
$\sin^{-1}(x)$	$[-1,1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$

Figure 7: The domain and range of $f(x) = \sin(x)$ and its inverse.

Roots

The roots of a function are the values of x such that $f(x) = 0$. A function can either real or complex (involving the imaginary unit i) roots. Indeed, an n-degree polynomial will have n zeroes with repeats allowed. Thus, an n-degree polynomial can be expressed as $P(x) = (x-r_1)(x-r_2)...(x-r_{n-1})(x-r_n)$, where the roots (real or complex) are $r_1, r_2, ..., r_{n-1}, r_n$.

To discuss root-finding with functions f, we will start with quadratics. Consider the quadratic $f(x) = x^2 - 4x + 3 = 0$. We can write f as $(x-3)(x-1) = 0$, and thus the roots are $x = 3$ and $x = 1$. Similarly, $g(x) = x^2 - 8x + 16$ can be reconfigured as $g(x) = (x-4)(x-4)$ so the duplicate root is $x = 4$. Essentially, the problem boils down to identifying the constants a, b, c, and d such that $f(x) = (ax+b)(cx+d)$. Then, f would have solutions $-\frac{b}{a}$ and $-\frac{d}{c}$. Expansion produces

$$f(x) = (ax+b)(cx+d) = acx^2 + (ad+bc)x + bd.$$

Thus, if the function was of the form $f(x) = a_2x^2 + a_1x^1 + a_0$, we would look for a, b, c, and d that enforce $ac = a_2$, $ad + bc = a_1$, and $bd = a_0$. A visual diagram of this strategy is provided.

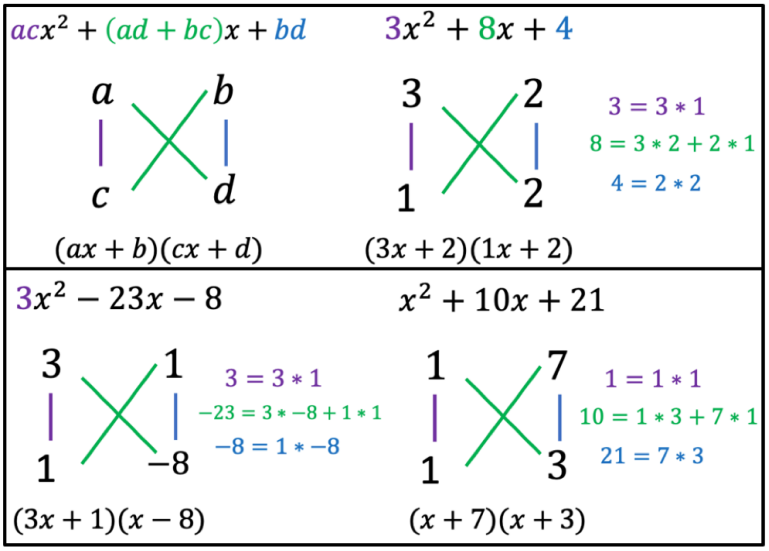


Figure 8: Transforming an expression of the form $a_2x^2 + a_1x^1 + a_0$ into a factored expression $(ax + b)(cx + d)$. Finding the right numbers is a process that involves both intuition and a degree of guess and checking.

We will implement the factoring technique to solve a contest problem.

Example 6

What is the product of all the roots of the equation (AMC 10B 2011)

$$\sqrt{5|x| + 8} = \sqrt{x^2 - 16}.$$

Solution: We take the square of both sides and solve for $|x|$.

$$5|x| + 8 = x^2 - 16$$
$$|x| = \frac{x^2 - 24}{5}.$$

Due to the absolute value, either $x = \frac{x^2-24}{5}$ or $x = -(\frac{x^2-24}{5}) = \frac{24-x^2}{5}$. In the first case, we have $5x = x^2 - 24$ or $x^2 - 5x - 24 = 0$. This factors into $(x-8)(x+3) = 0$ so $x = 8, 3$ are possible solutions.

In the second case, we have $5x = 24 - x^2$ so $x^2 + 5x - 24 = 0$. This factors into $(x+8)(x-3) = 0$ so $x = -8, 3$ are possible solutions. We need to identify which of the solutions actually mesh with the original equation since squaring both sides can easily generate extraneous solutions.

We note that plugging $x = \pm 3$ into the original equation leads to $\sqrt{5*3+8} = \sqrt{9-16}$ or $\sqrt{23} = \sqrt{-7}$, which is contradictory. We validate that $x = \pm 8$ leads to $\sqrt{5*8+8} = \sqrt{64-16}$ or $\sqrt{48} = \sqrt{48}$. Thus, the product of the solutions is $(8)(-8) = \boxed{\text{-64}}$.

In the next section, we will shine a lens into finding roots for polynomials beyond quadratics.

Rational Root Theorem

Rational Root Theorem

If polynomial with integer coefficients $f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x^1 + a_0 = 0$, where $a_0, a_n \neq 0$, has any rational solutions, they must be of the form $x = \frac{p}{q}$, where $\gcd(p, q) = 1$:

- p is a positive or negative integer factor of a_0.
- q is a positive or negative integer factor of a_n.

For instance, any rational root of the polynomial $x^3 + 2x^2 - x - 2$ must be among the set $\{\pm\frac{1}{1}, \pm\frac{2}{1}\}$. Similarly, consider the rational roots of the polynomial $6x^3 + x^2 - 5x - 2 = 0$. We consider the set of integer factors of $a_0 = -2$, or $\{\pm 1, \pm 2\}$, as well as the set of integer factors of $a_n = 6$, or $\{\pm 1, \pm 2, \pm 3, \pm 6\}$. All the rational roots of this polynomial can be attained by dividing one element from the integer factors of a_0 with one element from the integer factors of a_n. Thus, the set of possible rational roots is expounded below.

$$\{\pm\frac{1}{1}, \pm\frac{2}{1}, \pm\frac{1}{2}, \pm\frac{2}{2}, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{1}{6}, \pm\frac{2}{6}\} = \{\pm 1, \pm 2, \pm\frac{1}{2}, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{1}{6}\}$$

Searching through these carefully with some intuition, we notice that $x = 1$ works as $1^3 + 2*1^2 - 1 - 2 = 0$. Thus $x - 1$ must be a factor of the polynomial. Performing the long division of $\frac{6x^3+x^2-5x-2}{x-1}$ produces $6x^2 + 7x + 2$ so $6x^3 + x^2 - 5x - 2 = (x-1)(6x^2 + 7x + 2)$. We note that $6x^2 + 7x + 2$ can be factored into $(3x+2)(2x+1)$, so the other two solutions are $x = -\frac{2}{3}$ and $x = -\frac{1}{2}$.

Quadratic Equation

A canonical task is to find both solutions of a quadratic equation $f(x) = ax^2 + bx + c = 0$. The integer factoring approach discussed earlier works for a specific bandwidth of problems but the quadratic formula is universally applicable. The formula states that the solution to the equation $ax^2 + bx + c = 0$ can be expressed as the following.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Note that $f(x)$ will have two real solutions if the discriminant $b^2 - 4ac > 0$, a duplicate real solution if the discriminant is equal to 0, and two complex solutions with imaginary components if $b^2 - 4ac < 0$. For formality, a complex number is any number that can be expressed as $a + bi$, where a and b are real numbers and $i = \sqrt{-1}$ is the imaginary unit. Using this line of reasonining, we will attempt to dissect the following problem.

Example 7

There are two values of a for which the equation $4x^2 + ax + 8x + 9 = 0$ has only one solution for x. What is the sum of those values of a? (AMC 10A 2005)

Solution: We can rewrite the quadratic equation as $4x^2+(a+8)x+9=0$. We note that in order for the quadratic to have one solutions for x, the discriminant must be equal to 0. Thus, $b^2-4ac=0$, which means that $b^2-4ac=(a+8)^2-4(4)(9)=0$. Therefore, $a^2+16a+64-144=0$, which leads to $a^2+16a-80=0$. Using the quadratic formula,

$$x=\frac{-16\pm\sqrt{16^2-4(1)(-80)}}{2}=\frac{-16\pm\sqrt{576}}{2}=\frac{-16\pm 24}{2}.$$

Thus, x is either 4 or -20 and the sum of those values is $\boxed{\text{-16}}$.

Example 8

A real number a is chosen randomly and uniformly from the interval $[-20,18]$. The probability that the roots of the polynomial

$$x^4+2ax^3+(2a-2)x^2+(-4a+3)x-2$$

are all real can be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. (AIME II 6 2018)

Solution: We contemplate using Rational Root Theorem here. We note that possible rational roots to this polynomial include $x=\pm 1$ and $x=\pm 2$ since the last coefficient is 2 and the leading coefficient is 1. By plugging each of those values of x into the original expression, we find that $x=-2$ and $x=1$ collapse it into 0, as shown below.

$$\begin{aligned}1^4+2a(1)^3+(2a-2)*1^2+(-4a+3)*1-2 &= 1+2a+2a-2-4a+3-2\\ &= 0.\end{aligned}$$

$$\begin{aligned}(-2)^4+2a(-2)^3+(2a-2)*(-2)^2+(-4a+3)*(-2)-2 &= 16-16a+8a-8+8a-6-2\\ &= 0.\end{aligned}$$

Thus, the polynomial from above can be reconfigured as $(x-1)(x+2)R(x)$, where $R(x)$ is the quotient when the polynomial is divided by $(x-1)(x+2)=x^2+x-2$. Performing long division yields that $P(x)=x^2+(2a-1)x+1$. In order for the roots of the polynomial to all be real, $x^2+(2a-1)x+1$ must have real roots, which means that the discriminant must be at least 0. Thus, we can write $(2a-1)^2-4(1)(1)=4a^2-4a-3\geq 0$.

We can factor this expression into $(2a+1)(2a-3)\geq 0$, which means that either $a\geq\frac{3}{2}$ or $a\leq-\frac{1}{2}$. Since a is selected in the interval $[-20,18]$, the interval where a will result in exclusively real roots is $[-20,-\frac{1}{2}]\cup[\frac{3}{2},18]$. This interval has length 36 while the length of the total interval where a is selected along is 38. Thus, the probability of selecting an a within the desired interval is $\frac{36}{38}=\frac{18}{19}$. Therefore, $m=18$ and $n=19$ so $m+n=\boxed{37}$.

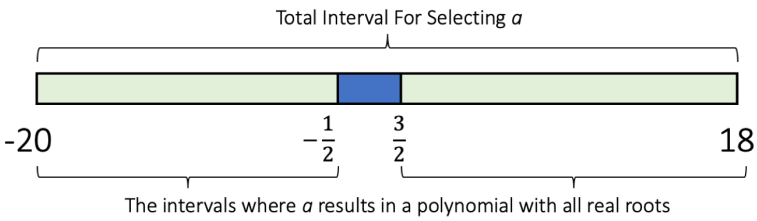


Figure 9: Accompanying Diagram for Example 10.

We will cover two more important principles with polynomials and functions.

Complex Conjugate Root Theorem

Let $P(x)$ be a polynomial function with real coefficients. If $a+bi$ is a root of P, where a and b are real numbers, then $a-bi$ must be a root of P as well.

We call $a-bi$ the *complex conjugate* of $a+bi$. In the problem below, we will demonstrate this theorem in action.

Example 9

The function $f(x) = x^3 + bx^2 + cx + 52$ has $\frac{13}{2-3i}$ as one of its zeros. Determine exactly the ordered pair (b, c). (MN Math League 2018/19)

Solution: We note that $\frac{13}{2-3i} = \frac{13}{2-3i} * \frac{2+3i}{2+3i} = \frac{26+39i}{4+9} = 2+3i$. According to the Complex Conjugate Root Theorem, $2-3i$ must also be a root of $f(x)$. Let a be the third root. Vieta's Formula states that the product of the roots is $-\frac{52}{1} = -52$, so $(2+3i)(2-3i)a = -52$. Thus, $(2^2-(3i)^2)a = 13a = -52$, so $a = -4$. Furthermore, we can use Vieta's Formula to unveil the values for b and c as follows, recalling that $-b$ is the sum of the roots and c is the sum of the roots *taken two at a time*.

$$-b = (2+3i) + (2-3i) + (-4) = 0$$
$$c = (2+3i)(2-3i) + (-4)(2+3i) + (-4)(2-3i) = 13 - 8 - 12i - 8 + 12i = -3.$$

Thus, the ordered pair (b, c) is $\boxed{\text{(0,-3)}}$.

Polynomial Remainder Theorem

Consider a polynomial function $P(x)$. The remainder of $P(x)$ divided by a linear polynomial $x-a$ is equal to $P(a)$.

Note that the remainder is equal to 0 whenever $P(a) = 0$, which indicates that $x-a$ is a factor of $P(x)$. To prove that this is true, suppose that we have have a polynomial $P(x)$ divided by the linear polynomial $x-a$. Let the quotient of this division be $Q(x)$ and let the remainder be $R(x)$. We can write $\frac{P(x)}{x-a} = Q(x) + \frac{R(x)}{x-a}$. Multiplying both sides by $x-a$ produces

$$P(x) = (x-a)Q(x) + R(x).$$

Plugging in $x = a$ will then produce

$$P(a) = (a-a)Q(a) + R(a) = R(a).$$

Thus, the remainder of the division is equal to $P(a)$, which completes the proof.

Exercises

1. Determine which of the following graphs are functions. Recall the vertical line test.
   - $y = x^2$
   - $x = 2$
   - $y = x^3 - x$
   - $\frac{(x-1)^2}{16} + \frac{(y-4)^2}{25} = 1$
2. At a store, when a length is reported as x inches that means the length is at least $x - 0.5$ inches and at most $x + 0.5$ inches. Suppose the dimensions of a rectangular tile are reported as 2 inches by 3 inches. In square inches, what is the minimum area for the rectangle? (AMC 10B 2011)
3. For a linear function of the form $f(x) = mx + b$, where m is the slope and b is the y-intercept, compute the inverse function $f^{-1}(x)$ in terms of x, m, and b. What is the relationship between f and f^{-1} if $m = 1$?
4. Let f be a linear function for which $f(6) - f(2) = 12$. What is $f(12) - f(2)$? (AMC 12B 2003)
5. What is the value of x if $|x - 1| = |x - 2|$? (AMC 10A 2004)
6. Suppose that a and b are nonzero real numbers, and that the equation $x^2 + ax + b = 0$ has solutions a and b. what is the pair (a, b)? (AMC 10B 2002)
7. For which of the following values of k does the equation $\frac{x-1}{x-2} = \frac{x-k}{x-6}$ have no solution for x? (AMC 10B 2002)
8. Find the value(s) of x such that $8xy - 12y + 2x - 3 = 0$ is true for all values of y. (AMC 10B 2002)
9. What is the sum of all possible values of k for which the polynomials $x^2 - 3x + 2$ and $x^2 - 5x + k$ have a root in common? (AMC 12A 2018)
10. Which of these describes the graph of $x^2(x + y + 1) = y^2(x + y + 1)$? (AMC 12A 2016)
    (a) two parallel lines
    (b) two intersecting lines
    (c) three lines that all pass through a common point
    (d) three lines that do not all pass through a common point
    (e) a line and a parabola
11. Find the remainder when $x^{13} + 1$ is divided by $x - 1$. (AHSME 1950)
12. For how many values of the coefficient a do the equations $0 = x^2 + ax + 1$ and $0 = x^2 - x - a$ have a common real solution? (AHSME 1977)
13. Let $f(x) = ax^7 + bx^3 + cx - 5$, where a, b, and c are constants. If $f(-7) = 7$, find $f(7)$. (ASHME 1982)
14. In multiplying two positive integers a and b, Ron reversed the digits of the two-digit number a. His erroneous product was 161. What is the correct value of the product of a and b? (AMC 10B 2011)
15. What is the sum of all integer solutions to $1 < (x - 2)^2 < 25$? (AMC 10B 2012)

16. Let P be a cubic polynomial with $P(0) = k$, $P(1) = 2k$, and $P(-1) = 3k$. What is $P(2)+P(-2)$ int erms of k? (AMC 12B 2014)

17. Let f be a function for which $f(\frac{x}{3}) = x^2 + x + 1$. Find the sum of all values of z for which $f(3z) = 7$. (AMC 12 2000)

18. Both roots of the quadratic equation $x^2 - 63x + k = 0$ are prime numbers. What is the number of possible values of k? (AMC 12A 2002)

19. Two lines with slopes $\frac{1}{2}$ and 2 intersect at $(2, 2)$. What is the area of the triangle enclosed by these two lines and the line $x + y = 10$? (AMC 10A 2019)

20. Let $P(x)$ be the polynomial $x^3 + Ax^2 + Bx + C$ for some constants A, B, and C. There exists constants D and E such that for all x, $P(x + 1) = x^3 + Dx^2 + 54x + 37$ and $P(x + 2) = x^3 + 26x^2 + Ex + 115$. Compute the ordered triple (A, B, C). (ARML 2016)

21. What is the least possible value of

$$(x + 1)(x + 2)(x + 3)(x + 4) + 2019$$

where x is a real number? (AMC 10A 2019)

22. The graph of $y = f(x)$, where $f(x)$ is a polynomial of degree 3, contains points $A(2, 4)$, $B(3, 9)$, and $C(4, 16)$. Lines AB, AC, and BC intersect the graph again at points D, E, and F, respectively, and the sum of the x-coordinates of D, E, and F is 24. What is $f(0)$? (AMC 12B 2017)

23. The real numbers c, b, a form an arithmetic sequence with $a \geq b \geq c \geq 0$. The quadratic $ax^2 + bx + c$ has exactly one root. What is this root? (AMC 10B 2013)

24. Let $f(x)$ be a third-degree polynomial with real coefficients satisfying

$$|f(1)| = |f(2)| = |f(3)| = |f(5)| = |f(6)| = |f(7)| = 12.$$

Find $|f(0)|$. (AIME I 2015)

25. Let m be the largest real solution to the equation

$$\frac{3}{x-3} + \frac{5}{x-5} + \frac{17}{x-17} + \frac{19}{x-19} = x^2 - 11x - 4$$

There are positive integers a, b, and c such that $m = a + \sqrt{b + \sqrt{c}}$. Find $a + b + c$. (AIME I 2014)